Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

Chapter 5 Large Number Law and Central Limit Theorem

Chebyshev's law of large numbers

Law of Large Numbers (LLN)

Let X_1, X_2... are sequence of independent random variables, they all have finite variance, and the variance has a common upper bound, that is, D(Xi) ≤ K, i =1, 2... , for any ε>0:

$$\lim_{n\to\infty} P\left\{\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \frac{1}{n}\sum_{i=1}^{n} E(X_i)\right| < \varepsilon\right\} = 1$$

Arithmetic mean

Mathematical expectation

- The deviation of the arithmetic mean and the mathematical expectation is small, with a probability close to 1.
- When *n* is sufficiently large, The probability arithmetic mean gets close to its mathematical expectation is close to 1.

Chebyshev's law of large numbers

Special case of Chebyshev's law of large numbers

Theorem(Weak Law of Large Numbers) If random variables $X_1, X_2, \cdots, X_n, \cdots$ are independent and with mean and finite variance

$$E(X_n) = \mu, Var(X_n) = \sigma^2, n = 1, 2, \cdots$$

then

$$\lim_{n\to\infty} \mathrm{P}\left(\left|\frac{1}{n}\sum_{k=1}^{n} X_k - \mu\right| < \varepsilon\right) = 1 \qquad \forall \varepsilon > 0$$

This LLN is known as **the weak law of large numbers**, **or the law of averages**. It says broadly that the average results of many independent random variables are stable and predictable.

Bernoulli's LLN

Theorem(Bernoullis LLN)

$$\lim_{n\to\infty} \mathrm{P}\left(\left|\frac{n_A}{n} - p\right| < \varepsilon\right) = 1 \qquad \forall \varepsilon > 0$$

- This means that the relative frequency n_A/n of the event A converges to the probability p of A with probability 1, i.e. n_A/n almost always converges to p.

- Therefore, in application, while the trial's times n is large enough, we can use the frequency to replace the probability of an event. This will help us to understand the precise meaning of frequency definition of probability.

- **Example:** If you toss a fair coin many, many times, the fraction of heads will approach 1/2.

Summary

- LLN describes the arithmetic mean and frequency stability of samples from various angles.
- Provides a theoretical basis for the methods commonly adopted by people:

- Replace or estimate the mean value with the arithmetic mean of the sample;
- To substitute or estimate its "probability" with frequency.

Chapter 5 Law of large numbers and Limit theorem

Law of Large Numbers (LLN)	Chebyshev's LLN
	Bernoullis's LLN
	Khinchin's LLN

Central Limit Theorem (CLT)	Lindberger-Levy CLT
	De Moirve-Laplace CLT
	Liapunov CLT

Introduction

- In probability theory, it has been known that normal distribution occupies the first important position, and many random variables obey normal distribution.

- Since Gauss pointed out that the measurement error obeys the normal distribution, it has been found that the normal distribution is very common in nature.

Gauss

Introduction

A large number of experimental observations indicate that:

If a **variable** is caused by the joint influence of a large number of independent and random factors, **and each individual factor plays little role** in the total influence, then this variable generally **obeys** or **approximately obeys** the normal distribution.

Is it based on experience, or **is it based on theory?**

Central Limit Theorems (CLT)

In probability theory, the distribution of the random variables sum converges to the normal distribution, which is called the central limit theorem.

The Central Limit Theorem (CLT) is a powerful result by a mathematician named George Polya (1887-1985) in 1920's. It is closely related to the LLN and it estimates the speed of convergence of the sample mean to the expected value.

- What is the limit distribution of this sum when n increases infinitely?
- Under what conditions will the limit distribution be normal?

Central Limit Theorems (CLT)

- The sum of a series of independent identically distributed random variables with finite variance, that it takes the standard normal distribution as limit after standardization.

  Lindberger-Levy CLT

  When the "same distribution" is a binomial distribution, a special case of the theorem is obtained.

  De Moirve-Laplace CLT

- For the variables caused and accumulated by a large number of small independent random factors (do not seek the same distribution), when the number of random factors tends to infinity, the normal distribution is the limit.

  Liapunov CLT

Lindberger-Levy CLT

Theorem Let $X_1, X_2, \cdots, X_n, \cdots$ be an independent and identically distributed sequence of random variables with finite mean μ and variance $\sigma^2 < \infty$, then

$$\lim_{n\to\infty} F_n(x) = \lim_{n\to\infty} P\left(\frac{\sum_{k=1}^{n} X_k - n\mu}{\sqrt{n}\sigma} \le x\right)$$

$$= \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt = \Phi\text{（x）}$$

where Φ(x) denotes the cdf of the standard normal distribution.

Lindberger-Levy CLT

Note： Lindberger-Levy CLT indicated that:

- When n is large enough, the sum of n independent identically distributed random variables with expectation and variance approximately obeys the standard normal distribution.
- Although it's hard to figure the exact distribution of $X_1 + X_2 + \ldots X_n$, but when n is large enough, the approximate distribution can be obtained.
- Although the distribution of random variables is arbitrary, as long as n is large enough, the distribution of sample mean value approximately obeys normal distribution.

$$\frac{1}{n}\sum_{k=1}^{n} X_k \sim N[E(X),(\sqrt{\frac{1}{n}}\sqrt{D(X)})^2]$$

$$\bar{X} \sim N(\mu, \sigma^2/n)$$

Lindberger-Levy CLT

Let $X = \sum_{k=1}^{n} X_k$ for all $n \geq 1$ and $\overline{X} = \frac{1}{n}\sum_{k=1}^{n} X_k$, then the following results are equivalent to CLT.

(1) $X \sim N\left(n\mu, n\sigma^2\right)$ approximately as $n \to \infty$.

(2) $\frac{X-n\mu}{\sqrt{n}\sigma} \sim \mathrm{N}(0,1)$ approximately as $n \to \infty$.

(3) $X \to Z \sim N\left(n\mu, n\sigma^2\right)$ as $n \to \infty$.

(4) $\lim_{n\to\infty} F_n(x) = \Phi(x)$,for all x, where $F_n(x)$ is cdf of $\frac{X-\mu}{\sqrt{n}\sigma}$.

(5) $\overline{X} \sim \mathbf{N}\left(\mu, \frac{\sigma^2}{n}\right)$ approximately as $n \to \infty$.

(6) $\frac{\overline{X}-\mu}{\left(\frac{\sigma}{\sqrt{n}}\right)} \sim \mathrm{N}(0,1)$ approximately as $n \to \infty$.

De Moivere-laplace CLT

Theorem

Let the random variables η_n be independent of each other and obey the binomial distribution with parameter n and p ($0 < p < 1$), then for any x there is always:

$$\lim_{n\to\infty} P\left(\frac{\eta_n - np}{\sqrt{np(1-p)}} \le x\right) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

De Moivere-laplace CLT

Note

- De Moivere-laplace CLT indicated that normal distribution is the limit distribution of binomial distribution. When n is large enough, the probability of binomial distribution can be calculated by normal distribution.

De Moivere-laplace CLT

- In Chapter 2, it has been introduced that when $\boldsymbol{n} \to \infty$, the binomial distribution takes Poisson distribution as the limit distribution; while in this chapter, the binomial distribution takes the normal distribution as the limit distribution.
- The difference between this two situations is that $\boldsymbol{np} \to \boldsymbol{\lambda}$ is required in Poisson's theorem (λ is a constant); while in the central limit theorem, $\boldsymbol{np} \to \infty$ is required.
- In practical calculation, if n is large but $\boldsymbol{np}$ or $\boldsymbol{nq}$ is small (i.e. $\boldsymbol{p}$ is small or $\boldsymbol{q=1-p}$ is small), we can use Poisson's theorem to approximate it; while $\boldsymbol{n}$, $\boldsymbol{np}$ or $\boldsymbol{nq}$ are large, then we can use the central limit theorem to approximate it.

Liapunov CLT

Theorem Let $X_1, X_2, \ldots, X_n, \ldots$ be an independent and

$$E(X_k)=\mu_k\ ,\ \ D(X_k)=\sigma_k^2\neq 0\ ,\ \ k=1,2\cdots$$

Let $B_n^2=\sum_{k=1}^{n}\sigma_k^2$, if there is an integer δ, that makes:

$$1\big/B_n^{2+\delta}\sum_{k=1}^{n}E\left\{\left|X_k-\mu_k\right|^{2+\delta}\right\}\to 0 \quad \text{when } n\to\infty\ .$$

Then, the CDF of ***Zn*** satisfy the follow formula for any x:

$$\lim_{n\to\infty}F_n(x)=\lim_{n\to\infty}P\left(\frac{\sum_{k=1}^{n}X_k-\sum_{k=1}^{n}\mu_k}{B_n}\le x\right)=\int_{-\infty}^{x}\frac{1}{\sqrt{2\pi}}e^{-\frac{t^2}{2}}dt$$

where, $Z_n=\dfrac{\sum_{k=1}^{n}X_k-E(\sum_{k=1}^{n}X_k)}{\sqrt{D(\sum_{k=1}^{n}X_k)}}=\dfrac{\sum_{k=1}^{n}X_k-\sum_{k=1}^{n}\mu_k}{B_n}$

Liapunov CLT

Note: Liapunov CLT indicated that:

- When n is large enough, $Z_n = \dfrac{\sum_{k=1}^{n} X_k - \sum_{k=1}^{n} \mu_k}{B_n}$ approximately obeys the standard normal distribution.

$$\sum_{k=1}^{n} X_k = B_n Z_n + \sum_{k=1}^{n} \mu_k \sim N\left(\sum_{k=1}^{n} \mu_k,\ B_n^{\ 2}\right)$$

- Liapunov CLT once again indicates the special position of normal distribution in probability theory. No matter what distribution each r.v.X obeys, as long as the conditions of Liapunov CLT are satisfied, their sum will approximately obey normal distribution when n is large enough.

Examples

Question: If more than 10 defective products are found during sampling inspection of product quality, this batch of products will be considered unacceptable.

Find: how many products should be inspected to make the probability of a batch of products with a defect rate of 10% unacceptable reach 0.9?

Solution: let n be the number of sampling products, X be the number of defective products. Then :

$$X \sim B(n, 0.1), \quad np = 0.1n, \quad \sqrt{npq} = 0.3\sqrt{n}$$

$$P(10 \le X \le n) = 0.9$$

$$\because \quad P(10 \le X \le n)$$

$$= P\left(\frac{10 - 0.1n}{0.3\sqrt{n}} \le \underline{\frac{X - 0.1n}{0.3\sqrt{n}}} \le \frac{n - 0.1n}{0.3\sqrt{n}}\right)$$

Examples

$$= P\left(\frac{10-0.1n}{0.3\sqrt{n}} \le \frac{X-0.1n}{0.3\sqrt{n}} \le 3\sqrt{n}\right)$$

"3σ rule"
$\Phi(3\sqrt{n}) = 1$

$$\approx \Phi(3\sqrt{n}) - \Phi\left(\frac{10-0.1n}{0.3\sqrt{n}}\right) = 1 - \Phi\left(\frac{10-0.1n}{0.3\sqrt{n}}\right)$$

$$\therefore \quad P(10 \le X \le n) = 1 - \Phi\left(\frac{10-0.1n}{0.3\sqrt{n}}\right) = 0.9$$

$$\frac{0.1n-10}{0.3\sqrt{n}} = 1.28 \quad \Rightarrow \quad n \approx 146$$

Examples

Question: When calculating addition, the computer takes each addend as the integer closest to it. Let all retrieval errors be independent random variables and obey uniform distribution in the interval [- 0.5,0.5].

Find: (1) The probability that the absolute value of the total error is less than 10 when the existing 1200 numbers are added;

(2) How many numbers should be added to make the absolute value of the error sum less than 10 and the probability greater than 0.9

Examples

Solution: Let X_1, X_2,... X_n be the retrieval error of each addend, then this is a column of independent and identically distributed random variables.

The error sum of all addends is : $\sum_{k=1}^{n} X_k$

$\because$ X_k $(k=1,2,\cdots n)$ obeys uniform distribution in $[-0.5,\ 0.5]$,

$$\therefore \quad E(X_k)=\frac{-0.5+0.5}{2}=0, \quad D(X_k)=\frac{[0.5-(-0.5)]^2}{12}=\frac{1}{12}$$

So, $\sum_{k=1}^{n} E(X_k)=0, \quad \sqrt{\sum_{k=1}^{n} D(X_k)}=\sqrt{\frac{n}{12}}$

Examples

Find: (1) The probability that the absolute value of the total error is less than 10 when the existing 1200 numbers are added;

(1) $\because n = 1200, \quad \therefore \sqrt{\sum_{k=1}^{n} D(X_k)} = \sqrt{\frac{1200}{12}} = 10$

Thus, $P\left(\left|\sum_{k=1}^{1200} X_k\right| < 10\right) = P\left(-10 < \sum_{k=1}^{1200} X_k < 10\right)$

$$= P\left(\frac{-10-0}{10} < \frac{\sum_{k=1}^{n} X_k - 0}{10} < \frac{10-0}{10}\right)$$

$$= P\left(-1 < \frac{\sum_{k=1}^{n} X_k}{10} < 1\right) \approx \Phi(1) - \Phi(-1)$$

$$= 2\Phi(1) - 1$$

$$= 2 \times 0.8453 - 1 = 0.6826$$

Examples

(2) How many numbers should be added to make the probability of the absolute value of sum error less than 10 is greater than 0.9?

$$P\left(\left|\sum_{k=1}^{n} X_k\right| < 10\right) > 0.9$$

Examples

$$\because\ P\left(\left|\sum_{k=1}^{n} X_k\right| < 10\right) = P\left(\frac{-10-0}{\sqrt{n/12}} < \frac{\sum_{k=1}^{n} X_k - 0}{\sqrt{n/12}} < \frac{10-0}{\sqrt{n/12}}\right)$$

$$= P\left(-20\sqrt{\frac{3}{n}} < \frac{\sum_{k=1}^{n} X_k}{\sqrt{\frac{n}{2}}} < 20\sqrt{\frac{3}{n}}\right) = 2\Phi\left(20\sqrt{\frac{3}{n}}\right) - 1$$

$$\Rightarrow\ P\left(\left|\sum_{k=1}^{n} X_k\right| < 10\right) > = 2\Phi\left(20\sqrt{\frac{3}{n}}\right) - 1 > 0.9$$

$$\therefore\ 20\sqrt{\frac{3}{n}} = 1.65 \quad \Rightarrow \quad n \approx 441$$